

# Classroom Experiments

Instructor guide: setting up the class, approving designs, running the live session and collecting the data

Groups of students design randomized experiments as surveys; on class day each group presents while everyone answers its survey on their phones and the results appear on the projected dashboard. This guide covers everything on the instructor side. Students have their own guide ( [docs/student-guide.pdf](#) ).

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## 1. Creating the classroom

On the site's home page, choose **Create Classroom** and give it:

- a **code** students type to join (e.g. `ECON101` ; unique, not case-sensitive);
- a **host password**, only for you: it opens the instructor pages;
- a **classroom password** that you share with the class (it's also in the QR code on the live dashboard);
- the **roster**: a CSV with the students' IDs and names. A Canvas gradebook export works as is.

Later, log in from **Host a Classroom** on the home page with the code and the host password. Students sign in with their SIS ID; the first time they confirm their first name and choose a password, so names and IDs in your data always match the roster.

## 2. The instructor home

Everything between classes is on the instructor home: the class phase, the roster, the surveys and the data.

**Start live session** opens the page you project in class.

### Behavioral Economics

Code **ECON101** · Student password **nudge**

[▶ Start live session](#)

#### Class phase

☒ **Building**

☐ Locked

☐ Feedback open

☐ Feedback closed

Next step: Locked →

☐ Lock survey edits & groups  
Students can still preview and download their responses.

☐ Accept comments  
Students can send and edit feedback.

☐ Show comments to groups  
Groups see the (anonymous) comments they received.

#### Peer feedback

Required comments per student  [Set](#)

Reviewers are assigned automatically when you lock surveys and groups. Students then see which groups they'll comment on.

#### Roster

**24 students, 20 signed up**

Max groups per student: 1 · max members per group: 5

[Manage roster](#)

#### Surveys

**6 surveys**

Max questions per survey  [Set](#)

Reminders and part 2 until  [Set](#)

Edit any survey, reorder groups, allow early deployment.

[Manage surveys](#) [Demo dashboard](#) [Images](#)

#### Data

**5 survey answers collected** ⓘ

[Download everything \(.zip\)](#)

Individual CSVs ▾

[▶ Danger zone](#)

- **Max questions per survey** (Surveys card): students can't save more; you're exempt.
- **Reminders and part 2 until**: the latest date any group's notifications or remote parts can fall on. Leave it empty for no limit.
- **Two-part dates and reminders to approve** lists groups waiting for you (see section 6).
- **Survey answers collected** counts each student who answered a survey once. It stays at 0 until surveys run (in class, launched on their own, opened early or a remote part), however many surveys groups have built.

### 3. The roster

**Manage roster** shows every student, whether they signed up, their group, its size and any pending invites. The filters at the top narrow the list: name or ID, signed up or not, in a group or not, with a pending invite, alone in their group, or group size fewer than, exactly or more than a number of members. "Showing X of Y" counts what's left.

Name or ID

Signed up or not

Not in a group

Group size

any

5

members

Clear

Showing 9 of 24

| Name             | SIS ID    | Status        | Group(s)                | Group size | Invites   | Actions                                     |
|------------------|-----------|---------------|-------------------------|------------|-----------|---------------------------------------------|
| Allen, Ella      | 912000021 | Not signed up |                         |            |           | <div>Remove</div>                           |
| Anderson, James  | 912000012 | Signed up     | 2026-10-09 03:15:20 UTC |            | 1 pending | <div>Reset password</div> <div>Remove</div> |
| Hall, Abigail    | 912000019 | Signed up     | 2026-10-09 03:15:21 UTC |            |           | <div>Reset password</div> <div>Remove</div> |
| Johnson, Ava     | 912000003 | Signed up     | 2026-10-09 03:15:19 UTC |            | 1 pending | <div>Reset password</div> <div>Remove</div> |
| King, Jack       | 912000022 | Not signed up |                         |            |           | <div>Remove</div>                           |
| Scott, Leo       | 912000024 | Not signed up |                         |            |           | <div>Remove</div>                           |
| Walker, Henry    | 912000018 | Signed up     | 2026-10-09 03:15:21 UTC |            |           | <div>Reset password</div> <div>Remove</div> |
| Wright, Scarlett | 912000023 | Not signed up |                         |            |           | <div>Remove</div>                           |
| Young, Daniel    | 912000020 | Signed up     | 2026-10-09 03:15:21 UTC |            |           | <div>Reset password</div> <div>Remove</div> |

Students not in a group yet.

- **Upload** again any time: new IDs are added, existing students keep their passwords and groups. Optionally remove students who aren't in the file.
- **Add one student** for late additions; **Reset password** if someone forgets theirs (they choose a new one on their next sign-in); **Remove** hides a student who dropped (their past answers stay in your data; you can restore them).
- **Max members per group** and **Max groups per student**: lowering them never removes anyone; full groups just can't take more.

### 4. Class phases

The **Class phase** card moves the class through four steps with **Next step**, or with individual switches:

| Phase           | What students can do                                                                                                                      |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Building        | Create groups, invite teammates, build and edit surveys.                                                                                  |
| Locked          | Surveys and groups are frozen (you can still edit). Each student sees the groups they'll comment on. Previews and downloads keep working. |
| Feedback open   | Students send their comments on other groups (each group opens for comments once its survey has run).                                     |
| Feedback closed | Comments are final; groups read the comments they received, anonymized.                                                                   |

Set **Required comments per student** before locking; assignments are balanced so every group gets about the same number of reviewers, never their own.

## 5. Managing surveys

**Manage surveys** lists every group's survey with its members (count, "alone" and "full" badges, pending invites), status and badges (not finished, two-part, waiting for your approval). Filter by title or member, by number of members, or by status.

### Surveys

Behavioral Economics (ECON101)

[+ New Survey](#)

Members any ▼

Any status ▼
[Clear](#)
Showing 6 of 6

| Group # | Title                                                           | Type                                        | Members                                                               | Early deployment            | Running it ⓘ                                                                                                                      | Actions                                        |
|---------|-----------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 1       | Anchoring and willingness to pay                                | <span>3 Q</span><br>numeric,multiple_choice | 2 <span>+1 invited</span><br>Martinez, Sofia; Chen, Liam              | <a href="#">Allow early</a> | in the live session<br><a href="#">Launch on its own</a>                                                                          | <a href="#">Edit</a><br><a href="#">Delete</a> |
| 2       | Hydration nudges<br><span>Two-part</span> <span>Approved</span> | <span>3 Q</span><br>numeric,multiple_choice | 3<br>Patel, Noah; Kim, Emma; Garcia, Mateo                            | <a href="#">Allow early</a> | <b>Part 1:</b> <span>✓ ran</span> <a href="#">Launch</a><br><b>Part 2:</b> remote, from the lobby<br><a href="#">Run again...</a> | <a href="#">Edit</a><br><a href="#">Delete</a> |
| 3       | Loss framing and donations                                      | <span>1 Q</span><br>multiple_choice         | 3<br>Nguyen, Olivia; Brown, Lucas; Lopez, Isabella                    | <a href="#">Allow early</a> | in the live session<br><a href="#">Launch on its own</a>                                                                          | <a href="#">Edit</a><br><a href="#">Delete</a> |
| 4       | Default effects in savings                                      | <span>1 Q</span><br>multiple_choice         | 2 <span>+1 invited</span><br>Wilson, Ethan; Singh, Mia                | <a href="#">Allow early</a> | in the live session<br><a href="#">Launch on its own</a>                                                                          | <a href="#">Edit</a><br><a href="#">Delete</a> |
| 5       | Social norms and recycling<br><span>Not finished</span>         | <span>1 Q</span><br>multiple_choice         | 1 <span>alone</span><br>Thomas, Amelia                                | <a href="#">Allow early</a> | in the live session<br><a href="#">Launch on its own</a>                                                                          | <a href="#">Edit</a><br><a href="#">Delete</a> |
| 6       | Decoy pricing for coffee                                        | <span>1 Q</span><br>multiple_choice         | 4<br>Hernandez, Benjamin; Moore, Harper; Jackson, Elijah; Lee, Evelyn | <a href="#">Allow early</a> | in the live session<br><a href="#">Launch on its own</a>                                                                          | <a href="#">Edit</a><br><a href="#">Delete</a> |

[Back to instructor home](#)

- **Edit** opens a group's survey page; you can always edit (saving a survey that has responses deletes them, after a warning).
- **Allow early** lets a one-part survey run before class: the group submits it with an answer-by date, classmates answer from their lobby, and it still runs in class. **Copy email text** gives you an announcement to paste.
- **Running it** shows how each survey (or each part) runs and whether it already ran:
  - **Launch** opens the live dashboard with only that survey or part, e.g. a part that runs at the start of class, or a group that presents on another day.
  - **Next Survey** in the live session skips everything that already ran.
  - **Run again...** (host password) deletes that survey's responses and notification sign-ups so it runs again as new, for a group that changed its survey after running it. Download its data first.

- **View survey** (from a survey's page or the lobby) shows all arms side by side; the flow chart shows each arm's path.

## 6. Two-part surveys and notifications

Both are optional; by default a survey has one part, no notifications, and runs in the live session. Groups set them up in two cards at the bottom of their survey page.

### Two-part surveys

Part 2 starts at a question the group chooses. Each part runs **in the live session, in class, launched separately** (by you, from Manage surveys), or **remote** (answered from the lobby between the group's dates). Part 2 is only for students who answered part 1, in the same arm; in the live session, others see "This survey is for classmates who answered its part 1" and wait for the next survey. The dashboard marks part-2 questions.

### Notifications

Per arm: a short note respondents read first, then messages, each with a time of day and chosen days. After answering (part 1), respondents add them to their calendar (iPhone and computers: a calendar file; Android: Google Calendar links) or turn on notifications from the site. Times are each phone's local time unless the group picks a time zone.

### Your approval

Notifications and remote dates need your approval: nothing is sent, and remote parts stay closed, until you approve. Review them on the group's **View survey** page, where they're listed under the questions with an **Approve** button. If the group changes them afterwards, they need approval again (the home page lists them).

Two parts and notifications

✓ Approved

Part 1: in class, during the live session. Part 2: remote, from the lobby, Wed Oct 7 to Sat Oct 17; everyone who answered part 1 gets a reminder when it opens (10 AM).

|                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Control</b></p> <p>You'll get a short reminder to stretch every morning this week.</p> <ul style="list-style-type: none"> <li>• <b>Time for a quick stretch!:</b> daily from Fri Oct 9 to Thu Oct 15 at 9 AM (7 times)</li> </ul> | <p><b>Nudge</b></p> <p>You'll get a notification about drinking water every day at noon this week, and two evening tips.</p> <ul style="list-style-type: none"> <li>• <b>Time for a glass of water!:</b> daily from Fri Oct 9 to Thu Oct 15 at 12 PM (7 times)</li> <li>• <b>Two liters a day keeps you sharp.:</b> every 3 days from Sun Oct 11 to Wed Oct 14 at 9 PM (2 times)</li> </ul> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Withdraw approval

Respondents then get no notifications, and remote parts close.

**Push notifications on the server** need nothing set up: the site creates its own keys on first use and sends due notifications every minute. Optionally set the environment variable `VAPID_SUBJECT` on Render to `mailto:` plus your email, as a contact for the push services. iPhones only get site notifications if the site is added to the Home Screen, so iPhone users are offered the calendar.

## 7. The demo dashboard

**Demo dashboard** (Surveys card on the home page) lets you click through every survey's questions, images and charts with made-up answers, without going live. Use it to check the surveys before class or to walk the class through the designs.

## Demo dashboard

Nothing goes live; answers are made up.

#1 Anchoring and willingness to pay

#2 Hydration nudges

#3 Loss framing and donations

#4 Default effects in savings

#5 Social norms and recycling

not finished

#6 Decoy pricing for coffee

All arms side by side

Flow chart

← Instructor home

**Demo** with **40** made-up respondents, so you can see how the results will look in class. These are random sample answers, not real data, and nothing is saved.

Sample size  Apply

The example file has the same made-up answers as the charts below, in exactly the format of your real download: one row per respondent, one column per question, plus time and timer columns.

Download example answers (CSV)

New random sample

# Anchoring and willingness to pay

40 responses / 40 students

Group #1

Q1: Willingness to pay

Q2: Would buy

Q3: Reason

### Response Distribution by Arm

| Value Range | Control (n=23) | High anchor (n=17) |
|-------------|----------------|--------------------|
| 18-26.3     | 0              | 3                  |
| 26.3-34.6   | 0              | 3                  |
| 34.6-42.9   | 2              | 7                  |
| 42.9-51.2   | 7              | 2                  |
| 51.2-59.5   | 4              | 1                  |
| 59.5-67.8   | 5              | 1                  |
| 67.8-76.1   | 4              | 0                  |
| 76.1-84.4   | 0              | 0                  |
| 84.4-92.7   | 0              | 0                  |
| 92.7-101    | 1              | 0                  |

| Arm         | N  | Mean  | Median | Std Dev | Min | Max |
|-------------|----|-------|--------|---------|-----|-----|
| Control     | 23 | 58.35 | 58     | 14      | 37  | 101 |
| High anchor | 17 | 37.53 | 38     | 10.5    | 18  | 60  |

## Questions by Arm

Control

High anchor

**Q1 (Willingness to pay):** How much would you pay for this mug (in \$)?

**Q2 (Would buy):** You said \${{Q1}}. Would you buy the mug at that price today?

**Q3 (Reason):** What made you hesitate?

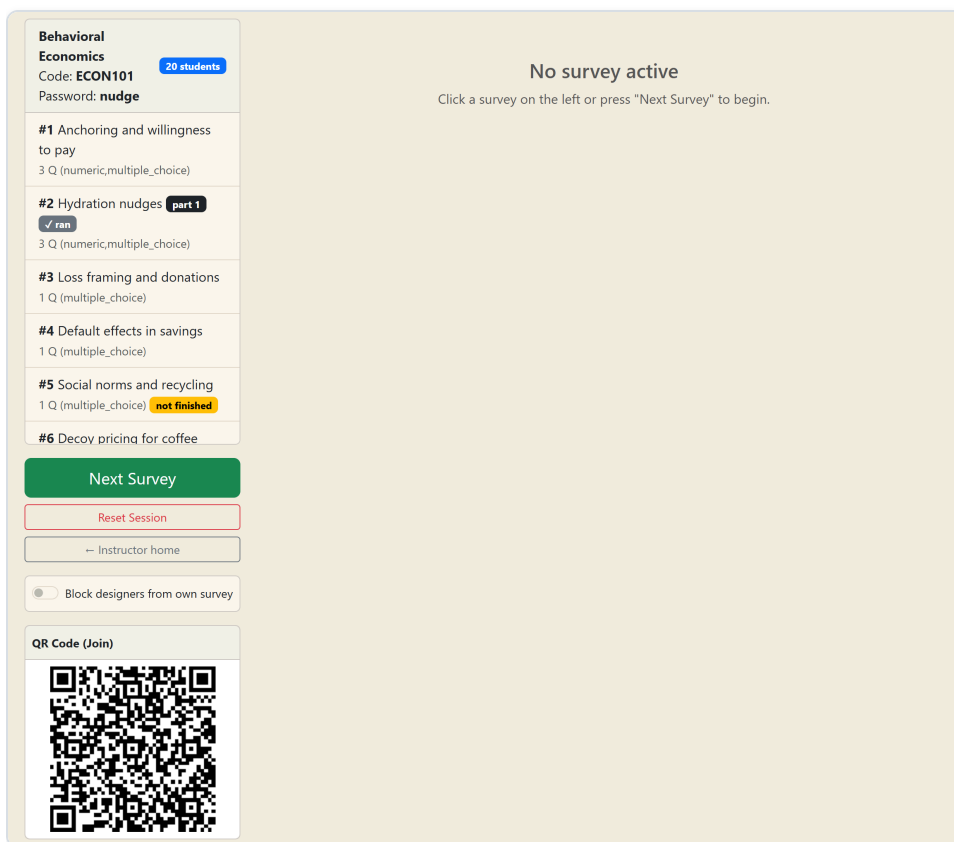
**Q1 (Willingness to pay):** Most students pay \$25 for this mug. How much would you pay (in \$)?

**Q2 (Would buy):** You said \${{Q1}}. Would you buy the mug at that price today?

**Q3 (Reason):** What made you hesitate?

## 8. Class day: the live session

1. On the home page, click **Start live session** and project it. Students join from their lobby, or by scanning the QR code with the classroom code and password.
2. Click a survey on the left, or **Next Survey**, to start it. Students' phones show it within a few seconds; each student always gets the same arm.
3. Click **Refresh Results** for the latest charts (they don't update on their own, to keep the site fast with a full class).
4. **Next Survey** moves on in group order, skipping surveys or parts that already ran; **Reset Session** stops the current survey (nothing is deleted).



- Two-part surveys show which part runs here ("part 2"); a "✓ ran" badge marks what already ran (click it to show its results again).
- **Block designers from own survey** shows group members "You designed this survey" instead of their own questions.
- Surveys run on another platform are listed for the order but don't start here.

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## 9. Peer feedback

When you lock surveys, each student is assigned groups to comment on. Comments are drafts until sent; students can jot private notes right after answering a live survey. Groups read the comments as "Reviewer 1, Reviewer 2"; you see names. **Feedback comments** and **Feedback summary** (required sent per student) are in the Data card for grading.

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## 10. Data and downloads

**Download everything (.zip)** on the home page has every file plus the uploaded images:

| File                                                 | Contents                                                                                                                               |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| all_responses.csv                                    | Every answer, with names, IDs, the respondent ID, arm, time on the question, left-page and timeout flags, and part.                    |
| surveys_config.csv                                   | What each arm showed: question texts and options.                                                                                      |
| participants.csv                                     | The roster key: names, IDs and the random respondent IDs students see.                                                                 |
| survey_designers.csv /<br>survey_participation.csv   | Who designed each survey; who answered each one.                                                                                       |
| feedback.csv /<br>feedback_summary.csv               | Comments and required-comment counts, for grading.                                                                                     |
| reminder_plans.csv                                   | Each group's notifications per arm, how its parts run, and approval.                                                                   |
| reminder_actions.csv /<br>reminder_notifications.csv | Who downloaded the calendar file, opened Google Calendar links, turned notifications on; each notification sent, delivered and tapped. |

Groups download their own survey's responses, anonymized (random respondent IDs only), in wide (one row per respondent) or long (one row per answer) format.

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## 11. Deploying and backups

- The site runs on Render with a persistent disk ( `/data` ) for the database and images. Pushing to the repository deploys it.
- Before deploying, download the zip, and if possible a copy of `/data/app.db` . Don't deploy during class: the site restarts.
- Updates only add to the database; existing classrooms, students, groups and surveys are kept.

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## 12. Troubleshooting

| Problem                               | What to do                                                                                                                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A student can't sign in               | Check they're on the roster (not removed); reset their password if needed.                                                                                                       |
| A survey doesn't appear on phones     | Students poll every 3 seconds; have them go back to the lobby and click Join. Check the survey isn't a part that's remote, or that it isn't already "ran" (click it to show it). |
| A group needs to run its survey again | Manage surveys → Run again... (download its data first).                                                                                                                         |
| A remote part doesn't open            | It needs your approval and its dates; see "to approve" on the home page.                                                                                                         |
| "0 survey answers collected"          | Normal before surveys run; it counts answers, not surveys built.                                                                                                                 |